

Unit 3 Classwork — Equations & Inequalities

One/Two-Step · Distributive · Both Sides · Multi-Step · Special Cases · Inequalities · Word Problems

A. One- & Two-Step Equations

Problem 1

Solve: $x + 9 = 15$

6

Problem 2

Solve: $4x = 28$

7

Problem 3

Solve: $3x + 5 = 20$

5

Problem 4

Solve: $x/3 - 2 = 4$

18

Problem 5

Solve: $-2x + 7 = 1$

3

B. Distributive Property & Variables on Both Sides

Problem 1

Solve: $2(x + 3) = 16$

5

Problem 2

Solve: $3(2x - 1) = 15$

3

Problem 3

Solve: $5x + 4 = 2x + 13$

3

Problem 4Solve: $7x - 2 = 3x + 14$ 4**C. Multi-Step Equations****Problem 1**Solve: $3(x + 2) = 2x + 10$ 4**Problem 2**Solve: $4x - 3 = 2(x + 5)$ 6.5**Problem 3**Solve: $5 - 2x = 3x + 15$ -2**D. Special Cases — No Solution or Infinite Solutions****Problem 1**Solve: $4x + 3 = 4x + 3$ — how many solutions?infinite**Problem 2**Solve: $2x + 5 = 2x - 1$ — how many solutions?no solution**Problem 3**Solve: $3(x + 4) = 3x + 12$ — how many solutions?infinite**E. Linear Inequalities****Problem 1**Solve: $x + 4 > 9$ $x > 5$

Problem 2

Solve: $3x \leq 21$

$$\underline{x \leq 7}$$

Problem 3

Solve: $-2x > 8$ (careful!)

$$\underline{x < -4}$$

Problem 4

Solve: $5x - 3 \geq 12$

$$\underline{x \geq 3}$$

Problem 5

Solve: $-3x + 1 \leq 10$ (careful!)

$$\underline{x \geq -3}$$

F. Word Problems**Problem 1**

A taxi charges \$3 plus \$2 per mile. If the total fare was \$17, how many miles were driven?

$$\underline{7}$$

Problem 2

A gym membership costs \$25 to join plus \$15/month. You can spend at most \$100. What's the max number of months?

$$\underline{5}$$